



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 2 - Day 6 Report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the work completed for Experiments 07 and 08 as part of the M-Tech Production & Marketing AI/ML Internship Program 2026. These experiments focused on two important areas of Machine Learning: Model Evaluation & Validation and Ensemble Learning Methods. During these experiments, I learned how to evaluate the performance of Machine Learning models using different validation techniques and performance metrics. I also explored advanced ensemble algorithms, including Random Forest, XGBoost, and LightGBM, and understood how combining multiple models can improve prediction accuracy, reliability, and overall performance.

2. Objectives

- Understand the importance of evaluating Machine Learning models before deployment.
 - Learn how to split datasets into training, validation, and testing sets.
 - Study evaluation metrics such as Accuracy, Precision, Recall, and F1-Score.
 - Understand overfitting, underfitting, and the bias-variance tradeoff.
 - Learn the concept of cross-validation and its importance in model evaluation.
 - Understand ROC Curves and AUC for classification models.
 - Explore the concept of ensemble learning.
 - Learn the difference between bagging and boosting techniques.
 - Compare the performance of Random Forest, XGBoost, and LightGBM algorithms.
 - Understand feature importance and model comparison techniques.
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3. Task Description

These experiments focused on evaluating Machine Learning models and improving prediction performance using ensemble learning techniques. I studied how different evaluation metrics measure model effectiveness and learned the importance of training, validation, and testing datasets in producing reliable models. I also explored cross-validation, ROC curves, and AUC values for classification problems. In addition, I studied ensemble learning algorithms, including Random Forest, XGBoost, and LightGBM, and learned how bagging and boosting techniques improve prediction accuracy by combining multiple models.

4. Work Done

During these experiments, I studied the complete process of evaluating Machine Learning models before deployment. I learned how datasets are divided into training, validation, and testing sets to ensure fair model evaluation and reliable performance. I practiced working with confusion matrices and gained an understanding of important evaluation metrics, including Accuracy, Precision, Recall, and F1-Score. I also studied the concepts of overfitting, underfitting, and the bias-variance tradeoff, which helped me understand how different models perform on training and unseen data. Furthermore, I explored cross-validation techniques to obtain more reliable evaluation results and studied ROC curves and AUC values to analyze classification model performance.

In the second experiment, I explored the concept of ensemble learning and understood how combining multiple Machine Learning models produces more accurate and stable predictions. I learned the differences between bagging and boosting techniques and studied the working principles of Random Forest, XGBoost, and LightGBM algorithms. I compared the performance of these models on sample datasets and analyzed feature importance to understand which input variables contributed most to the prediction process. These practical activities strengthened my understanding of advanced Machine Learning algorithms and their applications in solving real-world predictive problems.

5. Challenges Faced

Initially, understanding the differences between validation and testing datasets required careful study. Interpreting evaluation metrics such as Precision, Recall, F1-Score, ROC Curve, and AUC also required additional practice. In the ensemble learning section, differentiating between bagging and boosting techniques and understanding how hyperparameter tuning affects model performance were slightly challenging. However, after reviewing the concepts and completing the practical exercises, I developed a much clearer understanding of these topics.

6. Key Learnings

- Developed a strong understanding of Machine Learning model evaluation and validation techniques.
- Learned how training, validation, and testing datasets contribute to reliable model performance.
- Gained practical knowledge of Accuracy, Precision, Recall, F1-Score, ROC Curve, and AUC.
- Understood the concepts of overfitting, underfitting, and the bias-variance tradeoff.
- Learned how cross-validation improves the reliability of Machine Learning models.
- Studied the fundamentals of ensemble learning and its advantages.
- Understood the differences between Random Forest, XGBoost, and LightGBM algorithms.
- Learned the concepts of bagging, boosting, and feature importance.
- Improved my ability to evaluate, compare, and interpret the performance of multiple Machine Learning models.

7. Conclusion

Experiments 07 and 08 provided valuable practical knowledge in two essential areas of Machine Learning: model evaluation and ensemble learning. I developed a strong understanding of evaluating model performance using different validation techniques and performance metrics while also learning how ensemble algorithms improve prediction accuracy and reliability. These experiments enhanced my analytical skills and strengthened my practical understanding of advanced Machine Learning concepts, preparing me to build and evaluate more accurate and efficient predictive models in future AI and Machine Learning projects.

Intern Signature:



Supervisor:

Name:

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Name:

Date:

11 July 2026

Designation: